Recap

$V \cdot \vec{u} = \lambda \cdot \vec{u}$

$\vec{u}$ → ev.; $\vec{u}$ → ev.; λ → eigen value

eigen vectors are orthogonal

$V = \frac{X^T X}{N}$ → np.matmul (X^T, X) / len

V → np.cov

eg. 5 features

5 dim (features) $\xrightarrow{PCA}$ 5 PCs

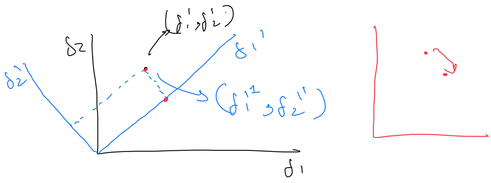
PC1 10
PC2 7
6
3
1

eigen values

explaine variance

5D → 3D

$\frac{10+7+6}{10+7+6+3+1} = \frac{23}{27} = 85\%$



eg. 5D

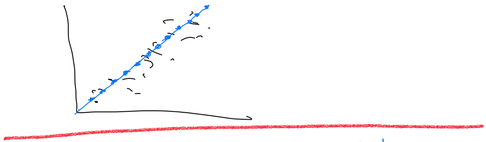
↳ 5 ev [ ]

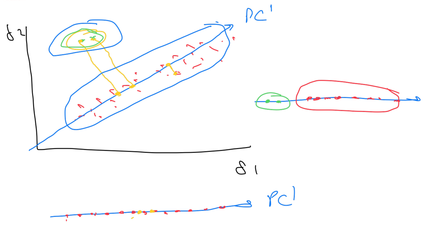
Inform d' explained var ratio $= \frac{\sum^{d} ev}{\sum ev}$

$d \rightarrow$ PCA → d PC

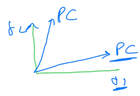
30% d ⇒ d'

preserved = 2?

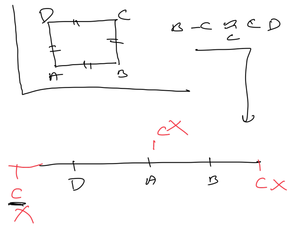
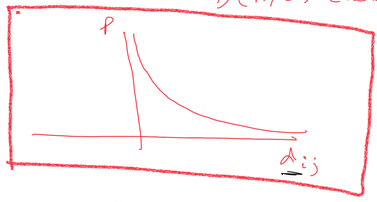
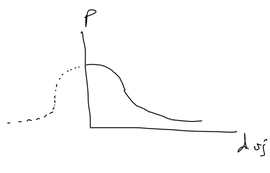




$x_i, x_j = \frac{1}{D(x_i, x_j)}$



Crowding Problem



$P(A,B) = \frac{1}{D(A,B)}$ euclid